

Leah Boustan

Data for “School Desegregation and Urban Change: Evidence from City Boundaries.”

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Methodology for Leah Platt Boustan’s Project

David Van Riper

Minnesota Population Center

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This document describes the GIS methodology developed for Leah’s dissertation research. It will contain the following sections: Goal, Inputs, Processing steps, Outputs.

Goal

The goal of this methodology is to create a table that shows the closest, defined as the straight-line distance, elementary and high school to each census tract in the sample. This table will allow Leah to attach the school characteristics to the census tracts.

Inputs

1. Address file for schools
2. Tract shapefiles
3. List of census tracts in the sample

Processing steps

1. Geocode the address file for schools. Outputs a point shapefile. (See “schoolmatching.txt” for details).
2. Append together the county-level tract shapefiles to make a single shapefile.
Shapefiles available at: <http://www.nhgis.org/>
Download only the counties that we used to geocode addresses.
3. Create a unique ID for every tract in the tract shapefile.
 - a. Concatenate the state, county, and aggtract attributes
4. Create a unique ID for every tract in the sample list.
 - a. Concatenate the state, county, and tract attributes (Leah handled this)
Sample tracts in tract_sample_1970_onlynew.xls
5. Define the projection of the schools shapefile. The projection needs to be the same as the projection of the streets data (Geographic coordinates, NAD 83)
6. Re-project the schools shapefile and the tracts shapefile to an equi-distance projection (US Contiguous Equidistant Conic – Equidistant Conic projection, meters, NAD 83). This type of projection conserves/maintains correct distances between two points.
7. Load the tract shapefile and the list of sample tracts into ArcMap and join the dataset on the unique ID. Select by Attributes and keep all tracts that matched.
8. Export the selected tracts to a new shapefile (just to reduce the number of tracts)
9. Calculate the centroid of every tract in the newly created shapefile.

- a. Add two new attributes to the polygon shapefile (X_Centroid and Y_Centroid). Need to be type Double, 19, 18.
 - b. Calculate values for the X_Centroid.
 - c. Calculate values for the Y_Centroid.
10. In Windows Explorer, create a copy of the census tracts's dBase file (the file must have the X and Y Centroids in it).
11. In ArcCatalog, right-click on the newly created dBase file and select "Create Feature Class" – "From XY Table". Fill in the proper sections of the form and create a point shapefile from the centroids.
12. If it doesn't have a projection, define the projection of the tract centroid point shapefile. It should be US Contiguous Equidistant Conic – equidistant conic projection, meters, NAD83).
13. In ArcMap, select all high schools from the schools shapefile and save as a new shapefile. This new shapefile should only contain high schools.
14. In ArcMap, select all elementary schools from the schools shapefile and save as a new shapefile. This new shapefile should only contain elementary schools.
15. In ArcMap, open the ArcToolbox and navigate to Analysis Tools → Proximity → Point Distance. Double-click on Point Distance to open the tool. This tool calculated the distance between a point in the Input Features to all points in the Near Features. Input Features are the sample tracts and near features are the schools, either elementary or high schools.
 - a. Put the sample tracts point shapefile in the Input Features box.
 - b. Put the high schools point shapefile in the Near Features box.
 - c. Choose a name and output location for the output table.
 - d. Set the search radius to 8046 meters (5 miles). This reduces the number of distance calculations.
 - e. Click OK to calculate the distances and create the table.
16. The output table from the Point Distance tool is the key to the rest of the processing. Thus, I want to discuss the schema of the table before continuing. The output table contains 4 fields – OID, Input_FID, Near_FID, Distance. Ignore the OID. **Input_FID** is the feature ID of the input feature (tract centroid). **Near_FID** is the feature ID of the near feature (school point). Distance is the straight-line distance between a tract and school, measured in meters.
17. In ArcMap, right-click on the Point Distance output table and select Joins and Relates → Join...
 - a. Join the output table to the sample tract point shapefile on Input_FID (output table) and FID (tract point shapefile).
 - b. Right-click on the point distance table again and select Joins and Relates → Join...
 - c. Join the output table to the high school shapefile on OUTPUT_TABLE_NAME.Near_FID and FID (high school shapefile).
18. In ArcMap, Open the output table (which will be large – lots of rows and columns because you have three tables joined together).
19. Click on the Options... button and select "Select by attributes..."
 - a. Select all rows where the tract jurisdictional code equals the school jurisdictional code.

20. Find the tract's unique ID column in the table and right click on it. Select Summarize... from the context menu.
 - a. For the high school distance item (high_tract_dist.DISTANCE), click the box next to MINIMUM. This will find the minimum distance for each tract.
 - b. Specify the output table and click OK.
21. The output from the Summarize function creates a table with the following attributes: OID, Tract Unique ID, Count_uniqueID, Minimum_DISTANCE. Ignore the OID and the Count_uniqueID. Minimum distance value is the shortest distance between a high school and the given census tract's centroid.
22. Right click on the Point Distance output table and select Joins and Relates → Joins..
 - a. Joins the Summarize... output table to the Point Distance output table on the sample tract unique ID.
23. Open the Point Distance output table (if it's not open already), and click on the Options... button and select "Select by attributes..."
 - a. Select all rows where Summary_table_Minimum_Distance = Point_Distance_output_table_Distance.
 - b. Click the Options... button and select Export... Export the selected records to a new dBase table.
 - c. The output of this function provides the information for the high school that is closest to the census tract, which is the goal of this GIS project.
24. Repeat steps 15-23 for the elementary school shapefile.

Outputs

1. A table where each row contains a census tract, the school closest to that census tract, and the distance between the tract centroid and the school.

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The original "mindist" function was not conducted over jurisdictions (because of the use of a 'select by attributes' rather than a 'definition query') so the tracts were matched to their closest school, regardless of the jurisdiction. That is, the matches did not follow school district lines.